

Project 1



Regression Task: Estimating Heart Rate (HR) and Heart Rate Variability (HRV) from Continuous PPG Signals

You are provided with:

1. **20 individual CSV files** (subject_1.csv to subject_20.csv), each containing continuous, clean, synthetic PPG signals for a unique subject.

Each file contains the following columns:

- time → Time in seconds
 - ppg_signal → PPG signal amplitude
 - subject_id → Subject identifier
2. A **master CSV file (master_data.csv)** containing the ground truth for each subject:
 - subject_id
 - hr (Heart Rate in bpm)
 - hrv (Heart Rate Variability in milliseconds, SDNN)



Your Task:

Develop a complete pipeline to estimate **Heart Rate (HR)** and **Heart Rate Variability (HRV)** for each subject using their continuous PPG signals.



Detailed Instructions:

1. **Preprocess the PPG signals**
 - Optionally apply denoising, baseline drift removal, or smoothing if necessary.
2. **Extract meaningful features** from each subject's PPG signal:
 - Time-domain features (mean, variance, peak counts, peak-to-peak intervals, etc.)
 - Frequency-domain features (if desired)
 - Derivative-based features or any other informative metrics you can derive
3. **Use the extracted features to build regression models** to predict:

- Heart Rate (HR)
 - Heart Rate Variability (HRV)
 - 4. You can use any regression technique:
 - Linear Regression
 - Random Forest Regression
 - Support Vector Regression (SVR)
 - Neural Networks
 - or Deep Learning models (CNN, LSTM, CNN-LSTM)
 - 5. **Evaluate your model's performance** using:
 - Mean Absolute Error (MAE)
 - Root Mean Square Error (RMSE)
 - R^2 Score
 - 6. **Compare the performance** of at least two different regression techniques.
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Deliverables:

- Clean, well-documented Python code / Jupyter Notebook
 - Preprocessed dataset or feature files (if generated separately)
 - Final regression models (or saved models if applicable)
 - A short report summarizing:
 - Preprocessing and feature extraction techniques used
 - Regression models tried and hyperparameter choices
 - Performance metrics for each model
 - Discussion on which approach worked best and why
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Bonus Task (Optional):

Create visualizations to show:

- A sample PPG waveform for any one subject
- Detected peaks and computed RR intervals

- HR and HRV estimation errors (actual vs predicted plots)
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✅ **Note:**

Use the **subject_id** column to map each subject's PPG data with their ground truth HR and HRV from the **master_data.csv**.

✅ **Dataset Link:**

[Dataset Link\(Click Here\)](#)